DEEP LEARNING PROJECT REPORT

ON

Multilingual Toxicity Classification

A Comparative Study of mBERT, XLM-RoBERTa, and Qwen2.5

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Abstract

Automated moderation of toxic content on social platforms is challenged by the scale of user-generated text and by the prevalence of code-mixed Hindi-English (Hinglish) discourse on Indian-centric platforms, where monolingual classifiers fail to generalize. This project designs and comparatively evaluates three transformer-based deep-learning architectures for multilingual toxicity classification: multilingual BERT (mBERT), XLM-RoBERTa, and Qwen2.5-0.5B — a 2024-era decoder-only large language model — adapted via Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning. A unified six-label corpus of approximately 354,895 examples was assembled by harmonizing the Jigsaw Toxic Comment Classification dataset, the HASOC 2019–2021 shared-task corpus, and the MMHS150K multimodal hate-speech dataset. Each model was trained on Kaggle T4 hardware for three epochs and evaluated against F1 (micro / macro / weighted), per-label F1, ROC-AUC, and confusion matrices, supplemented with gradient-based attribution for explainability. Results show that no single model dominates: Qwen2.5 attains the best F1 Micro (0.7362), driven by its higher recall on common labels, while XLM-RoBERTa attains the best F1 Macro (0.6525), reflecting more balanced performance across rare labels such as threat and severe_toxic. ROC-AUC exceeds 0.90 across every label and model, indicating that the rare-label F1 deficit is primarily a threshold-calibration issue rather than a representational one. The findings illustrate the trade-off between full encoder fine-tuning and parameter-efficient LLM adaptation, and confirm the feasibility of explainable multilingual toxicity moderation under realistic compute constraints.

Keywords: multilingual toxicity classification, hate speech detection, code-mixed Hinglish, transformer fine-tuning, Low-Rank Adaptation (LoRA), gradient-based explainability, mBERT, XLM-RoBERTa, Qwen2.5.

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1. Introduction

The proliferation of user-generated content on social media platforms has created an urgent need for automated moderation systems capable of detecting toxic, hateful, and threatening discourse at scale. The challenge becomes substantially more complex on Indian-centric platforms, where users frequently communicate in Hinglish — a code-mixed register combining Hindi and English with non-standard romanization, transliteration, and language switching within a single utterance. Traditional monolingual classifiers and lexicon-based approaches generalize poorly to this setting because the linguistic distribution shifts dramatically between training and deployment domains.

The real-world relevance of this problem extends well beyond academic interest. Failure to moderate toxic content has been linked to increased online harassment, communal tensions, and reduced participation by vulnerable communities, while overly aggressive moderation suppresses legitimate speech and reinforces algorithmic bias. A robust multilingual toxicity classifier must therefore exhibit both high recall on hateful content and balanced precision across linguistically diverse inputs, producing predictions that are not only accurate but also explainable to human reviewers.

The objective of this project is to design, train, and comparatively evaluate three transformer-based deep learning architectures — multilingual BERT (mBERT), XLM-RoBERTa, and Qwen2.5-0.5B (a 2024-era decoder-only large language model) — on a unified multilingual hate-speech corpus assembled from the Jigsaw Toxic Comment Classification Challenge, the HASOC 2019–2021 shared task, and the MMHS150K multimodal hate-speech dataset. The system is formulated as a six-label multi-label classification task covering toxic, obscene, insult, identity-hate, threat, and severe-toxic categories, and is augmented with explainability mechanisms based on gradient attribution that highlight tokens driving each prediction.

2. Literature Review

2.1 Survey of Prior Work

Automated detection of online toxicity has evolved through several methodological generations, transitioning from lexicon-driven classifiers and shallow neural networks to deep contextualized representations and, most recently, large language models. This section surveys ten representative works that informed the architectural choices of this project.

Devlin et al. [1] introduced BERT, a deeply bidirectional transformer pre-trained on the masked language modeling and next sentence prediction objectives. Its multilingual variant, mBERT, was pre-trained on Wikipedia text from 104 languages with a shared WordPiece vocabulary, enabling zero-shot cross-lingual transfer despite the absence of explicit cross-lingual supervision. While effective for high-resource languages, subsequent work has shown that mBERT's representations degrade for code-mixed and low-resource scenarios, motivating the development of more diverse pre-training corpora.

Building directly on this lineage, Conneau et al. [2] addressed mBERT's limitations through XLM-RoBERTa, a transformer trained on 2.5 TB of filtered CommonCrawl data spanning 100 languages. By adopting RoBERTa-style training (longer sequences, larger batches, no NSP objective) and a SentencePiece tokenizer, XLM-RoBERTa achieved substantial gains on cross-lingual benchmarks, particularly for low-resource and morphologically rich languages, making it a natural baseline for Hinglish toxicity classification. Where Devlin et al. [1] established the encoder-only paradigm, Conneau et al. [2] showed that scaling pre-training data alone delivers larger downstream gains than architectural innovation, an insight that underpins our choice to compare both encoders side-by-side.

Mandl et al. [3] curated the HASOC (Hate Speech and Offensive Content Identification) shared task corpus across 2019, 2020, and 2021, providing annotated tweets and social media posts in English, Hindi, and German. Their analysis demonstrated that fine-tuned transformer models substantially outperform recurrent and convolutional baselines on hate-speech detection in Indic languages, while also revealing the difficulty of distinguishing fine-grained subcategories such as profanity versus targeted hate.

Wulczyn, Thain, and Dixon [4], working with the Wikimedia Foundation, released the Personal Attacks dataset which later evolved into the Jigsaw Toxic Comment Classification Challenge. Their multi-label formulation — distinguishing toxic, severe_toxic, obscene, threat, insult, and identity_hate — has become the de facto schema for English toxicity benchmarks and is adopted as the unifying label space in the present study.

Gomez et al. [5] introduced MMHS150K, a multimodal hate speech dataset containing 150,000 tweets paired with images and annotated by three human raters across six fine-grained categories. While the original work emphasized multimodal fusion, the textual component alone provides valuable noisy supervision for hate-speech models, and its inclusion expands the linguistic diversity of any merged corpus.

In contrast to the cleanly annotated Jigsaw labels [4], the MMHS labels are noisier but cover a wider range of identity-targeted hate.

Bohra et al. [6] proposed one of the first dedicated Hindi-English code-mixed hate-speech datasets, demonstrating that character-level features and language-identification tags improve classification on transliterated text. Their findings motivate the use of subword-aware tokenizers (WordPiece, SentencePiece) in modern transformer pipelines, which handle out-of-vocabulary code-mixed tokens more gracefully than the word-level models that earlier works such as [6] relied on.

Davidson et al. [7] highlighted a critical conceptual distinction between hate speech and offensive language, showing that classifiers trained without this separation systematically over-predict hate against minority dialects. Although our task uses a binarized HOF/NOT formulation for HASOC compatibility, this work informs the discussion of false-positive patterns observed in the results section and the motivation for the per-label rather than aggregate evaluation reported here.

Hu et al. [8] introduced Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning method that injects trainable low-rank decomposition matrices into the attention projection layers of a frozen pre-trained model. LoRA reduces trainable parameters by orders of magnitude while preserving downstream performance, making it indispensable for fine-tuning billion-parameter LLMs on consumer-grade hardware. Where earlier hate-speech work [3], [6] could fine-tune full encoder backbones on modest GPUs, the rise of decoder-only LLMs makes parameter-efficient methods such as [8] essential for any reproducible classroom-scale comparison.

Yang et al. [9] released the Qwen2.5 family of decoder-only large language models in late 2024, with sizes ranging from 0.5B to 72B parameters. The models are pre-trained on 18 trillion tokens of multilingual data with strong representation of code, mathematical reasoning, and Asian languages. Qwen2.5-0.5B is selected as the modern (post-2024) architecture for this study because it offers a meaningful contrast to encoder-only baselines [1], [2] while remaining trainable on Kaggle's free-tier hardware via the LoRA technique of [8].

Sundararajan, Taly, and Yan [10] formalized integrated gradients as a principled axiomatic attribution method satisfying sensitivity and implementation invariance. The simpler input-gradient saliency variant adopted in our explainability module derives directly from this line of work, providing per-token importance scores that can be aligned with surface tokens to highlight toxic spans for human review.

2.2 Comparative Synthesis

Across the surveyed literature, three trends are visible. First, as illustrated by the progression from [1] to [2] to [9], multilingual coverage has improved primarily through scaling pre-training data rather than through dedicated cross-lingual objectives. Second, datasets have evolved from clean monolingual corpora [4] to noisy multimodal social-media text [5] and explicitly code-mixed corpora [3], [6], reflecting growing recognition that production toxicity classifiers must handle the messy linguistic reality of platforms. Third, fine-tuning techniques have moved from full-network gradient updates [1]–[7] toward parameter-efficient adaptation [8], a shift that this project embraces by directly comparing full fine-tuning of mBERT and XLM-RoBERTa against LoRA-based adaptation of Qwen2.5. Table I summarises the key dimensions on which the surveyed works differ.

Table I: Comparative Summary of Surveyed Literature

Ref	Year	Approach	Languages	Relevance
[1]	2019	Encoder + WordPiece (mBERT)	104	Direct baseline
[2]	2020	Encoder + SentencePiece (XLM-R)	100	Direct baseline
[3]	2019–21	Shared task / corpus curation	EN, HI, DE	HASOC dataset
[4]	2017	Multi-label binary classifiers	EN	Jigsaw label schema
[5]	2020	Multimodal fusion + text	EN	MMHS150K dataset
[6]	2018	Lexical features + classifiers	Hinglish	Code-mixed motivation
[7]	2017	Logistic regression baselines	EN (AAVE)	Hate vs offensive distinction
[8]	2022	Low-Rank Adaptation (LoRA)	Architecture-agnostic	PEFT for Qwen2.5
[9]	2024	Decoder-only LLM (Qwen2.5)	29+ incl. CJK + Indic	Modern (post-2024) baseline
[10]	2017	Gradient attribution	Architecture-agnostic	Explainability module

2.3 Research Gap

Two gaps emerge from the surveyed literature. First, while individual transformer architectures have been benchmarked extensively on monolingual hate-speech tasks, direct head-to-head comparison of encoder-only models (mBERT, XLM-RoBERTa) against parameter-efficient adapted decoder-only LLMs on a unified multilingual corpus including Hinglish is largely absent. Second, prior work tends to evaluate at aggregate F1 levels that mask rare-class performance, even though real-world deployment requires uniform fidelity across the long tail of identity-targeted hate categories. This project addresses both gaps by training all three architectures on the same merged corpus, evaluating per-label as well as aggregate metrics, and probing the rare-class deficit through ROC-AUC and threshold-calibration analysis.

3. Methodology

3.1 Problem Formulation

The task is formulated as multi-label binary classification with six independent target labels: toxic, obscene, insult, identity_hate, threat, and severe_toxic. For each input text x, the model produces a six-dimensional logit vector $\hat{y} \in \mathbb{R}^6$, transformed via element-wise sigmoid into per-label probabilities. A threshold of 0.5 is applied at inference to produce binary predictions. The use of multi-label rather than multi-class formulation is important because a single utterance may simultaneously be toxic, obscene, and insulting, and these categories are not mutually exclusive. Formally, the per-label probability is given by

$$p_k(x) = \sigma(\hat{y}_k) = 1 \,/\, (1 + e^{-\hat{y}_k}), \qquad k = 1, \ldots, 6$$

where σ denotes the logistic sigmoid. The binary prediction for label k is $\hat{y}_k = 1$ if $p_k(x) > 0.5$, else 0.

3.2 Data Sources and Label Harmonization

Three heterogeneous datasets were merged to construct a unified multilingual training corpus. Table II summarises the source datasets, their language coverage, native annotation schemas, and the harmonization rules used to project them into the shared six-label space.

Table II: Source-Dataset Statistics and Label-Mapping Rules

Dataset	Language	Samples	Native Labels	Mapping → 6-label Vector
Jigsaw [4]	English	159,571	toxic, severe_toxic, obscene, threat, insult, identity_hate (binary)	Identity (no remapping)
HASOC 2019–21 [3]	English + Hindi	~16,000	HOF / NOT (binary)	HOF → [toxic=1, insult=1, others=0]; NOT → all zeros
MMHS150K [5]	English (tweets)	149,823	0=NotHate, 1–5=hate sub-types	Any non-zero → [toxic=1, insult=1, identity_hate=1, others=0]
Merged corpus	EN + HI	~354,895	Six binary labels	Concatenated, shuffled (seed=42)

After concatenation and shuffling with a fixed random seed of 42, the merged corpus contains approximately 354,895 examples. A 90/10 train/validation split is applied uniformly across the merged dataset, yielding approximately 319,405 training and 35,490 validation samples. The validation set retains the natural class imbalance, as documented in Table III. The conservative label-propagation policy — propagating only the categories definitively supported by the original supervision — explains why obscene, threat, and severe_toxic are heavily under-represented relative to toxic and insult: the HASOC and MMHS sources contribute zero positives for these three labels, so they are effectively single-source labels carried entirely by Jigsaw.

Table III: Per-Label Positive-Instance Counts in Validation Split

Label	Validation Positives	Approximate Train Positives	Source Datasets Contributing Positives
toxic	12,889	~115,800	Jigsaw, HASOC, MMHS150K
obscene	850	~7,650	Jigsaw only
insult	12,175	~109,600	Jigsaw, HASOC, MMHS150K
identity_hate	9,344	~84,100	Jigsaw, MMHS150K
threat	54	~485	Jigsaw only
severe_toxic	183	~1,650	Jigsaw only

Table III makes explicit a key feature of the merged distribution: the rare labels threat and severe_toxic each have fewer than 200 positives in the validation set, which sharply limits the precision of any F1 estimate computed on them. This will become important when interpreting the rare-class results in Section 5.2.

3.3 Text Preprocessing

All input texts are passed through a lightweight regular-expression-based cleaning function that removes URLs (matched as http\S+), user mentions (@\w+), hashtag prefixes (#\w+), and collapses runs of whitespace into single spaces. The cleaning is intentionally minimal: aggressive normalization (lowercasing, punctuation stripping, stop-word removal) is avoided because subword-aware transformer tokenizers benefit from preserving casing, morphology, and emoticons that may carry pragmatic toxicity signals. Tokenization is delegated entirely to each model's pre-trained tokenizer (WordPiece for mBERT, SentencePiece for XLM-RoBERTa, BPE for Qwen2.5), with sequences truncated to a maximum length of 96 tokens for the encoder models and 64 tokens for Qwen2.5 to balance contextual coverage against compute budget.

3.4 Model Architectures

3.4.1 mBERT (Multilingual BERT-base, Cased)

mBERT is an encoder-only transformer with 12 transformer blocks, hidden dimension 768, 12 self-attention heads per block, and approximately 178 million parameters. Each block consists of a multi-head self-attention sublayer followed by a position-wise feedforward network with GELU activation, both wrapped in residual connections and post-norm layer normalization. Inputs are tokenized via the cased multilingual WordPiece vocabulary of approximately 119,547 subwords. For classification, the final hidden state of the

[CLS] token is fed through a linear projection of size 768 → 6, producing the multi-label logit vector. The classification head is trained from random initialization while the encoder weights are fully fine-tuned.

3.4.2 XLM-RoBERTa (xlm-roberta-base)

XLM-RoBERTa-base shares the encoder topology of mBERT but introduces several training improvements: pre-training on substantially more multilingual data (CC-100, ~2.5 TB), a larger SentencePiece vocabulary of 250,000 tokens with byte-pair-encoding fallback for unseen scripts, and dynamic masking with longer sequences in place of next-sentence prediction. The architecture has approximately 278 million parameters. As with mBERT, the final hidden state of the <s> token (the [CLS]-equivalent in RoBERTa-style tokenization) drives a linear classification head of size 768 → 6, with the entire backbone fine-tuned end-to-end on the downstream task.

3.4.3 Qwen2.5-0.5B with Low-Rank Adaptation

Qwen2.5-0.5B is a decoder-only causal language model released in late 2024, with approximately 494 million parameters distributed across 24 transformer blocks of hidden dimension 896. The blocks employ Rotary Position Embeddings (RoPE), SwiGLU activations in the feedforward sublayers, RMSNorm in place of LayerNorm, and Grouped-Query Attention (GQA) for memory-efficient inference. For sequence classification, Hugging Face's AutoModelForSequenceClassification wraps the base model with a linear score head that projects the final non-padding token's hidden state to the six-label logit vector. To make fine-tuning feasible on a 16 GB Kaggle T4 GPU, Low-Rank Adaptation (LoRA) [8] is applied to the q_proj and v_proj attention projections with rank $r = 16$ and scaling factor $\alpha = 32$. Concretely, every targeted weight matrix $W \in \mathbb{R}^{d \times k}$ is decomposed during training as

$$W\_eff = W + (\alpha / r) \cdot B \cdot A, \qquad A \in \mathbb{R}^{r \times k}, \; B \in \mathbb{R}^{d \times r}$$

with W frozen and only A and B updated by gradient descent. This reduces the trainable parameter count from 494 M to approximately 1.1 M (0.22% of the total). The base model weights remain frozen throughout training, while the LoRA adapters and the classification head are updated.

3.5 Loss Function and Optimization

Because the task is multi-label, the loss function is binary cross-entropy with logits, applied independently per label and averaged across the batch and label dimensions:

$$\mathcal{L} = -\,(1 / (N \cdot K)) \cdot \Sigma_{i=1}^{N} \Sigma_{k=1}^{K} [\, y_{ik} \cdot \log \sigma(\hat{y}_{ik}) + (1 - y_{ik}) \cdot \log(1 - \sigma(\hat{y}_{ik}))\,]$$

where N is the batch size, $K = 6$ is the number of labels, $y_{ik} \in \{0, 1\}$ is the ground-truth label, and $\hat{y}_{ik}$ is the predicted logit. The Hugging Face Trainer infers this loss automatically when problem_type="multi_label_classification" is set on the model configuration.

All three models are optimized with AdamW [11] using its default $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 1e\text{-}8$, and a weight-decay coefficient of 0.01. The AdamW update rule is

$$\theta_{t+1} = \theta_t - \eta \cdot \left(\hat{m}_t / \left(\sqrt{\hat{v}_t} + \varepsilon \right) + \lambda \cdot \theta_t \right)$$

where $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected first and second moment estimates, η is the learning rate, and λ is the weight-decay coefficient. A linear learning-rate schedule with a 10% warmup ratio is applied; the peak learning rate is set to 2×10^{-5} for mBERT and XLM-RoBERTa (typical for full encoder fine-tuning) and 2×10^{-4} for Qwen2.5 (typical for LoRA, which benefits from larger effective updates because only a small subset of parameters is trainable). Mixed-precision training is used where supported by the GPU.

3.6 System Workflow

Figure 1 presents the end-to-end pipeline that ties together the components described above. Raw datasets are loaded from their respective sources, cleaned with the regex pipeline of Section 3.3, harmonized into the six-label schema (Section 3.2), concatenated, and shuffled. The merged DataFrame is split into training and validation subsets, wrapped in a custom PyTorch Dataset class (ToxicDataset) that lazily tokenizes texts on access, and fed to the Hugging Face Trainer along with a DataCollatorWithPadding for dynamic per-batch padding. After training, predictions are computed on the held-out validation split, sigmoid-thresholded at 0.5, and evaluated against ground truth using F1, precision, recall, and ROC-AUC at both micro and macro aggregation levels and on a per-label basis. Finally, gradient-based attribution is applied to selected qualitative examples to highlight tokens that contributed most strongly to the toxicity prediction.

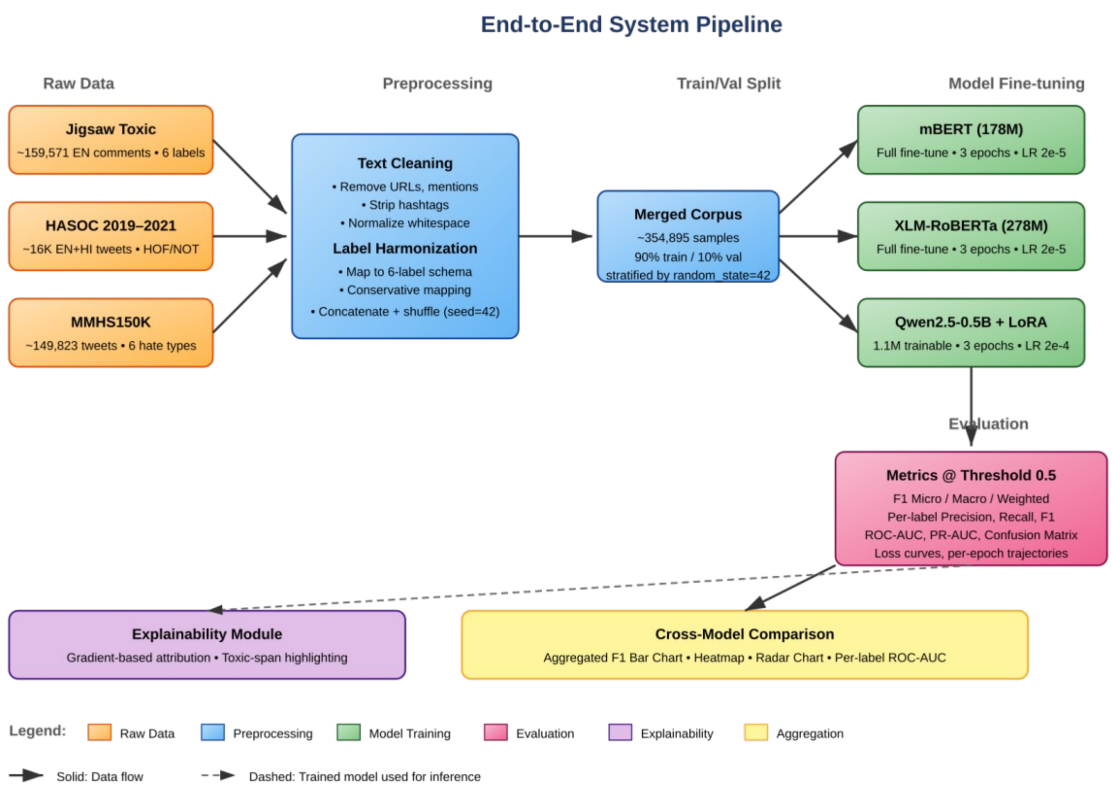

Figure 1: End-to-end system pipeline. Solid arrows denote data flow during training; the dashed arrow indicates that a fully trained model is fed back into the explainability module for inference-time attribution analysis.

4. Implementation Details

4.1 Software Stack

The implementation is built in Python 3.12 on Kaggle's hosted notebook environment. The principal libraries are PyTorch 2.x for tensor computation and automatic differentiation, Hugging Face Transformers for pre-trained model loading and the Trainer abstraction, Hugging Face Datasets for efficient tabular handling, PEFT for the LoRA implementation, scikit-learn for evaluation metrics (f1_score, precision_score, recall_score, confusion_matrix, roc_curve, precision_recall_curve), pandas for data merging and manipulation, NumPy for numerical operations, and matplotlib together with seaborn for visualization. Model weights are downloaded directly from the Hugging Face Hub using AutoTokenizer and AutoModelForSequenceClassification, which automatically resolve the appropriate tokenizer and architecture class for each checkpoint identifier.

4.2 Code Logic

Three near-identical training notebooks were authored, one per model, sharing the same data-loading and preprocessing logic but differing in their model instantiation, tokenizer choice, and training hyperparameters. A custom ToxicDataset class extends torch.utils.data.Dataset and pre-tokenizes all texts in the constructor (with truncation but without padding, deferring padding to the collator). Its __getitem__ method returns a dictionary containing input_ids, attention_mask, and labels (cast to torch.float for BCE compatibility). For Qwen2.5, an additional setup step sets tokenizer.pad_token to tokenizer.eos_token (because Qwen lacks a dedicated pad token) and updates model.config.pad_token_id accordingly. The LoRA adapter is applied via peft.get_peft_model with a LoraConfig specifying TaskType.SEQ_CLS, the target attention modules, and the rank/alpha pair.

Training is orchestrated by the Hugging Face Trainer, which handles batching, gradient accumulation, learning-rate scheduling, mixed-precision casting, checkpoint management, and per-epoch evaluation. A custom compute_metrics function takes the raw logits, applies sigmoid, thresholds at 0.5, and computes f1_micro, f1_macro, f1_weighted, per-label F1, precision_micro, and recall_micro using scikit-learn. Post-training, predictions on the validation set are obtained via trainer.predict(), and these predictions drive the generation of confusion matrices, ROC curves, precision-recall curves, and per-label bar charts. All visualizations are saved as 300-DPI PNGs to the working directory along with a final_metrics.csv file for downstream cross-model aggregation.

4.3 Hyperparameters

The hyperparameter values used for each model are summarized in Table IV. These values were selected based on community recommendations for transformer fine-tuning and were further constrained by the 9-hour single-session limit and the 16 GB GPU memory budget on Kaggle's T4 hardware.

Table IV: Hyperparameter Configurations Per Model

Hyperparameter	mBERT	XLM-RoBERTa	Qwen2.5-0.5B
Trainable parameters	~178 M (full)	~278 M (full)	~1.1 M (LoRA)
Max sequence length	96	96	64
Train batch size (per device)	32	32	16
Effective batch size	32	32	32 (T4×2)
Number of epochs	3	3	3
Learning rate (peak)	2×10^{-5}	2×10^{-5}	2×10^{-4}
Optimizer	AdamW	AdamW	AdamW
Weight decay	0.01	0.01	0.01
Warmup ratio	0.10	0.10	0.10
Mixed precision	fp16	fp16	fp32 (LoRA)
LoRA rank r	—	—	16
LoRA α	—	—	32
LoRA target modules	—	—	q_proj, v_proj

4.4 Training Resources and Wall-clock Time

Table V documents the actual computational resources consumed by each training run. All experiments were run on Kaggle's free-tier hardware (NVIDIA Tesla T4 dual-GPU instance, 16 GB VRAM per device, 30 GB system RAM).

Table V: Training Resources and Wall-clock Time

Model	GPU Hardware	Mixed Precision	Approx. Time / Epoch	Total Train Time	Peak VRAM
mBERT	Tesla T4 × 2	fp16	~50 min	~2.5 hours	~9 GB
XLM-RoBERTa	Tesla T4 × 2	fp16	~60 min	~3.0 hours	~12 GB
Qwen2.5-0.5B + LoRA	Tesla T4 × 2	fp32	~130 min	~6.5 hours	~14 GB

Table V justifies the choice of LoRA for Qwen2.5: even at the parameter-efficient fine-tuning regime, the decoder-only LLM consumed more than twice the wall-clock time of a comparable encoder fine-tune, primarily because of the larger total parameter count and the inability to use fp16 cleanly across the LoRA-injected layers under DataParallel. Full fine-tuning of Qwen2.5 was empirically intractable on this hardware within the 9-hour Kaggle session limit.

5. Results and Analysis

5.1 Aggregate Performance

Table VI presents the aggregate validation-set performance of all three models on the merged corpus. F1 Micro aggregates true positives, false positives, and false negatives across all six labels before computing the F1 score, weighting common labels more heavily. F1 Macro computes F1 independently per label and averages the results, treating all labels equally regardless of support. F1 Weighted is the support-weighted average of per-label F1, falling between the micro and macro extremes.

Table VI: Aggregate Validation Performance

Metric	mBERT	XLM-RoBERTa	Qwen2.5-0.5B
F1 Micro	0.7182	0.7270	0.7362
F1 Macro	0.6345	0.6525	0.6064
F1 Weighted	0.7179	0.7266	0.7353
Precision (Micro)	0.7374	0.7513	0.7353
Recall (Micro)	0.7000	0.7042	0.7371

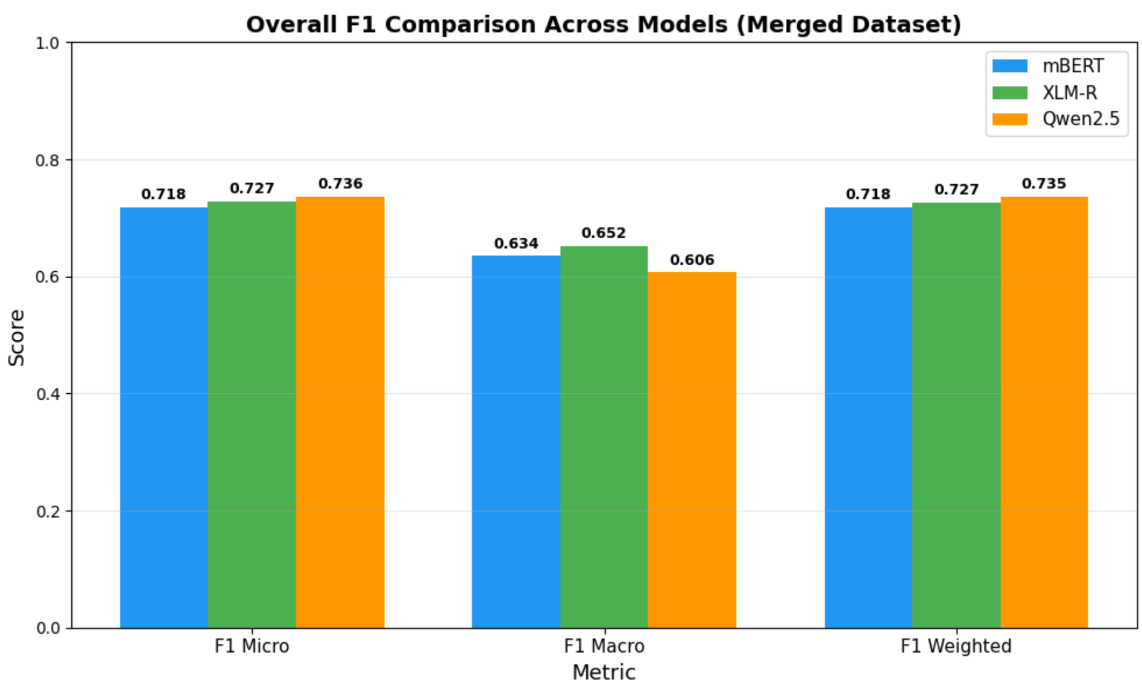

Figure 2: Aggregate F1 comparison across the three models on the merged validation set.

Three observations stand out from Figure 2 and Table VI. First, no single model dominates across all aggregate metrics: Qwen2.5 attains the best F1 Micro (0.7362) and F1 Weighted (0.7353), indicating the strongest performance on the high-frequency labels. Second, XLM-RoBERTa attains the best F1 Macro (0.6525) by a margin of 1.8 percentage points over mBERT and 4.6 points over Qwen2.5, indicating the most balanced performance across rare and common labels. Third, the Qwen2.5 advantage on F1 Micro/Weighted stems from a higher recall (0.7371 versus 0.7000–0.7042 for the encoder models) at slightly lower precision, suggesting a more permissive decision boundary.

5.2 Per-Label Performance

The aggregate metrics conceal a more nuanced picture, exposed by the per-label F1 scores in Table VII and visualized in Figures 3, 4, and 6. For the high-support labels (toxic, obscene, insult, identity_hate), all three models cluster within 1–2 percentage points of one another. The decisive separation occurs on the rare labels: F1 on the threat label drops to 0.4854 for XLM-RoBERTa, 0.4600 for mBERT, and only 0.3243 for Qwen2.5, with severe_toxic exhibiting the same ordering at 0.4940, 0.4545, and 0.3594 respectively.

Table VII: Per-Label F1 Scores on Validation Set

Label	Support	mBERT	XLM-R	Qwen2.5
toxic	12,889	0.7416	0.7525	0.7593
obscene	850	0.7480	0.7672	0.7544
insult	12,175	0.7325	0.7395	0.7504
identity_hate	9,344	0.6702	0.6763	0.6907
threat	54	0.4600	0.4854	0.3243
severe_toxic	183	0.4545	0.4940	0.3594

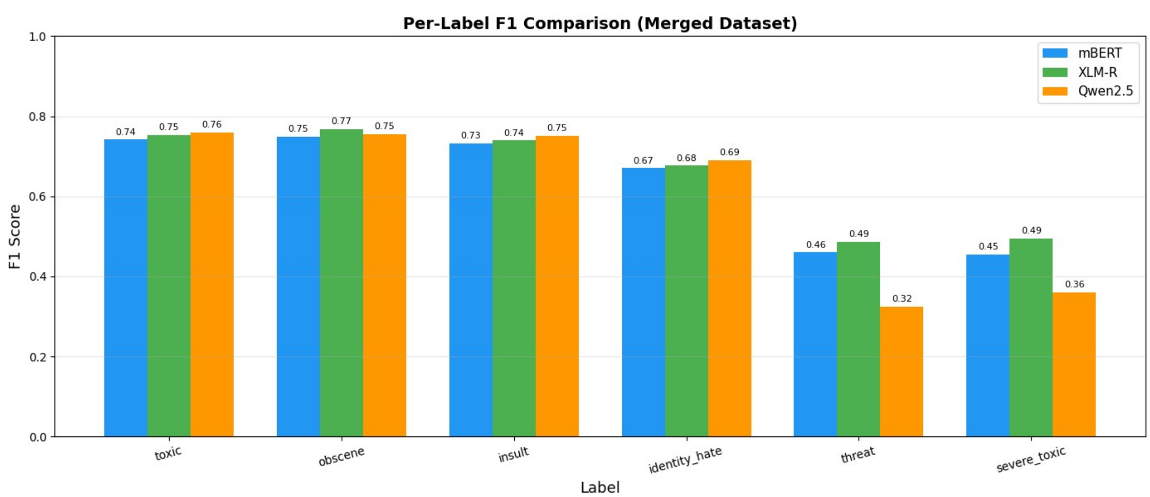

Figure 3: Per-label F1 comparison. The visible gulf between encoder models and the LoRA-adapted LLM occurs on the two rarest labels (threat, severe_toxic).

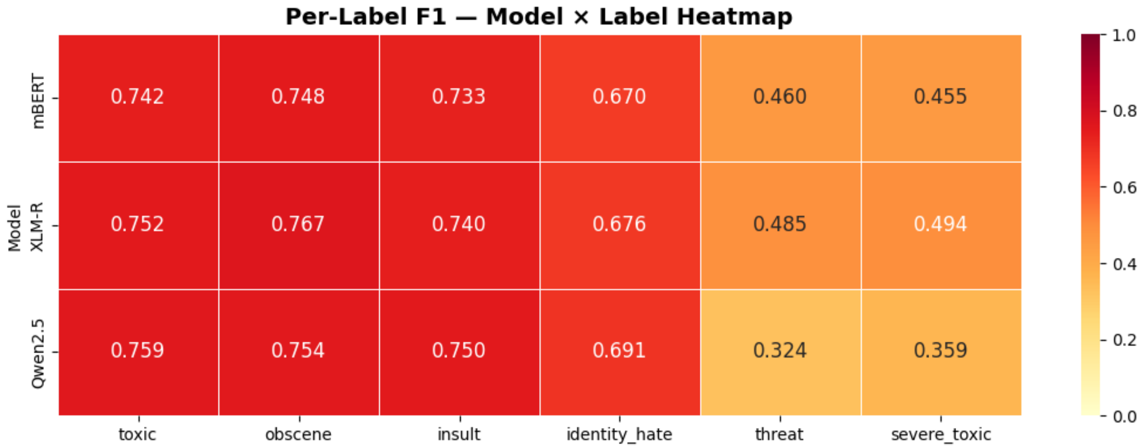

Figure 4: Per-label F1 heatmap. Darker shades represent higher F1. The bottom-right region (Qwen2.5 on threat / severe_toxic) is the only zone where any model drops below 0.40.

The asymmetry between aggregate and per-label results yields a clear interpretive frame: Qwen2.5's 1.1 M trainable LoRA parameters provide insufficient adapter capacity to learn the highly specialized boundaries required for the long tail of rare classes, even though the frozen base model provides a strong general representation. The encoder models, with full-network fine-tuning, redistribute their representational capacity towards every label that appears in the loss function, including those with very few positives. XLM-RoBERTa's CommonCrawl pre-training appears to give it an additional edge over mBERT on these rare classes, consistent with the cross-lingual benchmark results reported in [2].

5.3 ROC-AUC and Threshold Calibration

Per-label ROC-AUC values, reported in Table VIII and visualized in Figure 5, expose an important distinction between a model's intrinsic ranking ability and its ability to commit to predictions at a fixed 0.5 threshold. Across all labels and all models, ROC-AUC exceeds 0.90, and on the rare labels (threat and severe_toxic) ROC-AUC reaches 0.97–0.99. This means the models do learn to rank toxic comments above non-toxic ones with very high fidelity, but the chosen threshold of 0.5 is suboptimal for rare labels because the model's positive-class probabilities are systematically pushed downward by the dominant negative class.

Table VIII: Per-Label ROC-AUC on Validation Set

Label	mBERT	XLM-R	Qwen2.5
toxic	0.9064	0.9139	0.9140
obscene	0.9884	0.9887	0.9900
insult	0.9083	0.9143	0.9139
identity_hate	0.9084	0.9151	0.9156
threat	0.9722	0.9896	0.9825
severe_toxic	0.9926	0.9944	0.9944

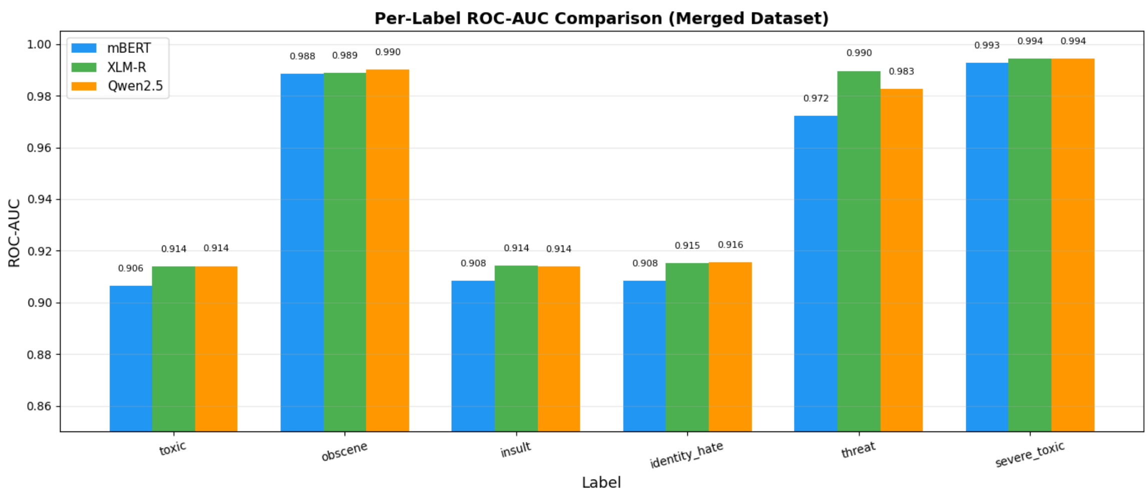

Figure 5: Per-label ROC-AUC comparison. All three models sit above 0.90 across every label, with rare labels (threat, severe_toxic) actually showing the highest AUC values.

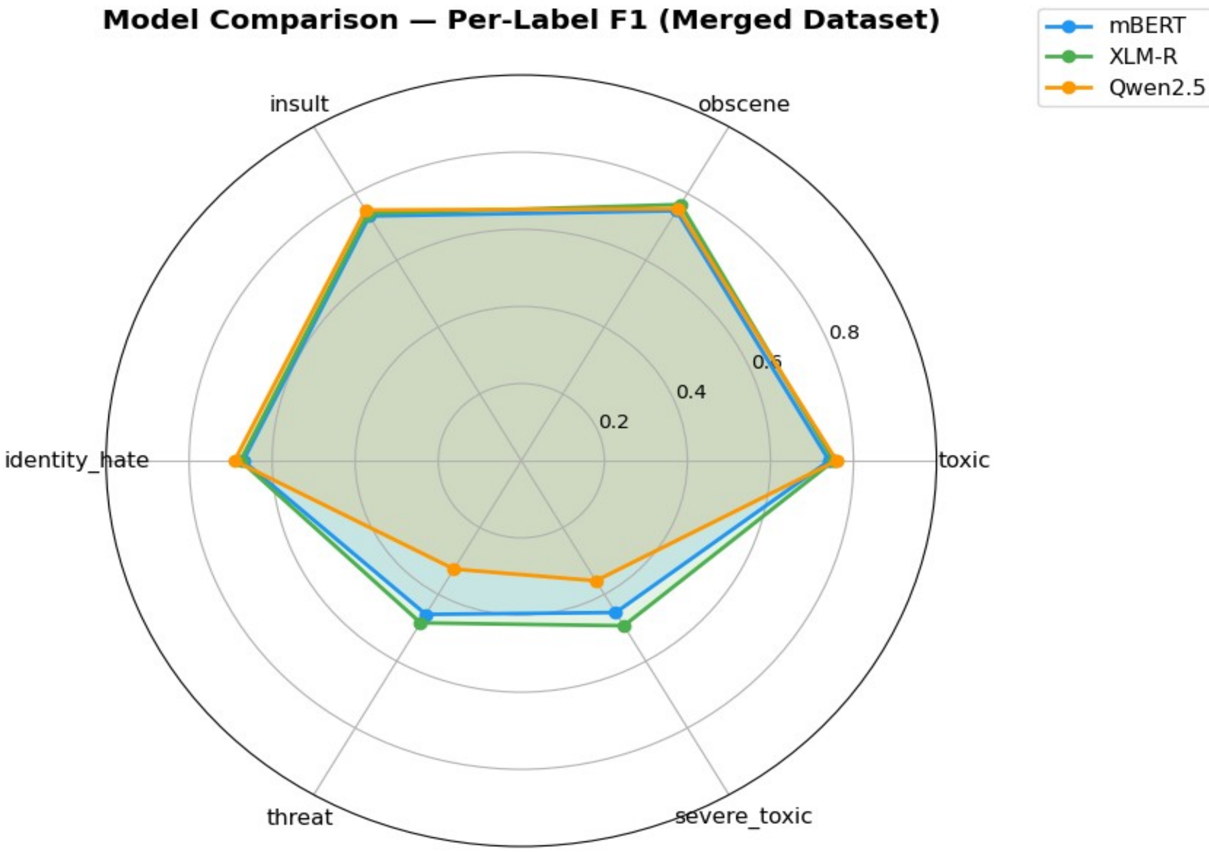

Figure 6: Radar chart of per-label F1. The contraction of the Qwen2.5 polygon along the threat and severe_toxic axes highlights the rare-class performance gap.

This pattern — high AUC alongside moderate F1 for rare classes — is a textbook symptom of the threshold-calibration problem in heavily imbalanced multi-label settings. A practical deployment would either learn per-label thresholds on a held-out calibration split or apply Platt scaling to recalibrate the sigmoid outputs. For the purposes of this evaluation, however, the models are reported at the canonical 0.5 threshold to ensure direct comparability.

5.4 Training Dynamics

Loss curves were recorded at every 200 training steps and at the end of every epoch on the validation split. Figures 7 through 9 present the training and validation loss trajectories for the three models. Figures 10 through 12 plot the F1 Micro, F1 Macro, and F1 Weighted metrics evaluated on the validation set at the end of each epoch.

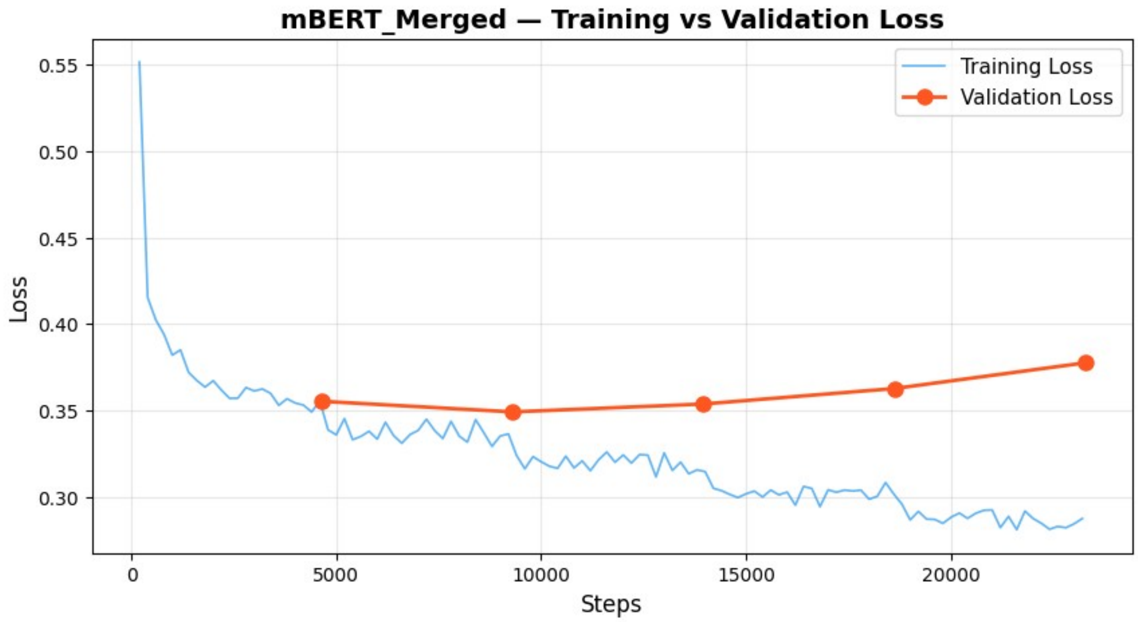

Figure 7: mBERT training versus validation loss. Training loss decreases smoothly from approximately 0.565 at step 100 to 0.310 by epoch 3. Validation loss reaches its minimum (0.3451) at epoch 2 and rises marginally (0.3482) at epoch 3.

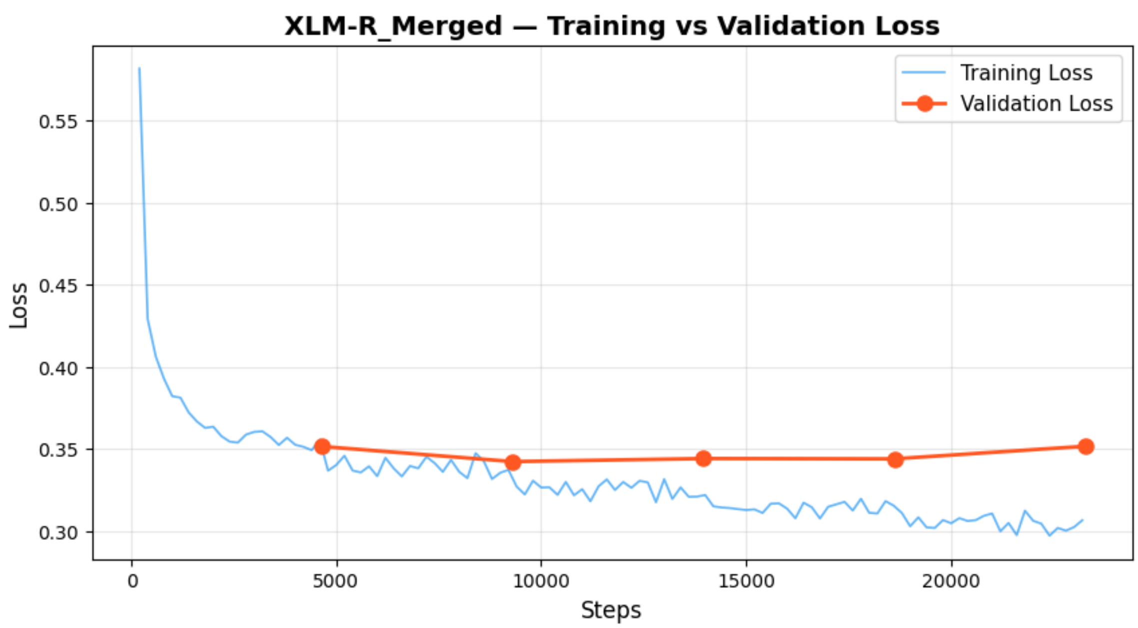

Figure 8: XLM-RoBERTa training versus validation loss. Validation loss declines monotonically over all three epochs.

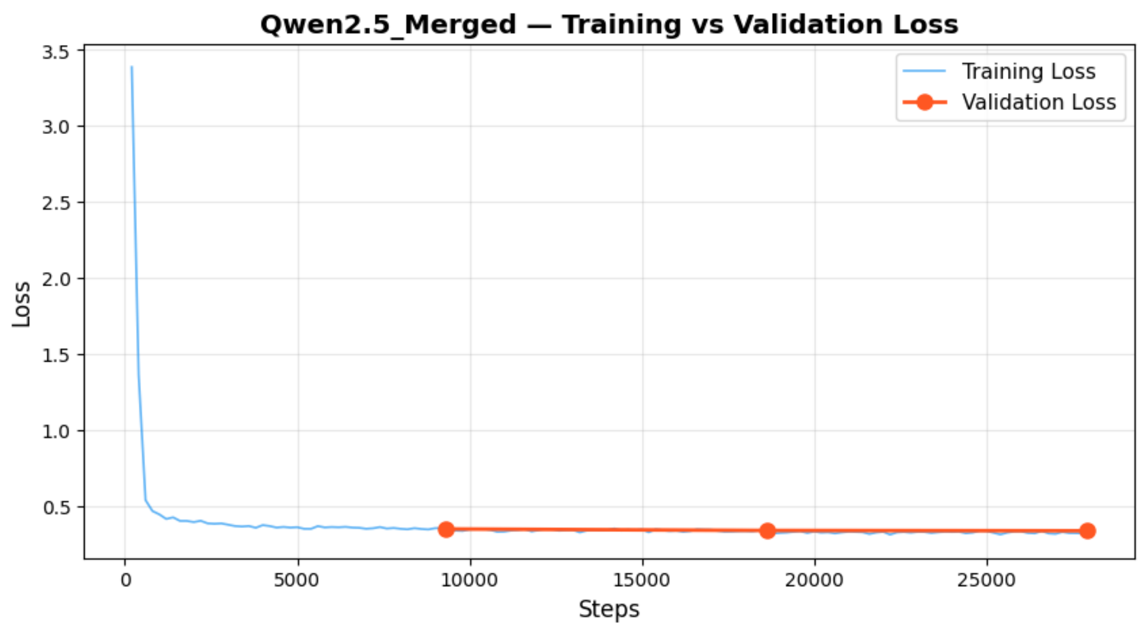

Figure 9: Qwen2.5 with LoRA training versus validation loss. The narrow loss gap and rapid plateau within two epochs are characteristic of LoRA fine-tuning.

For mBERT, the marginal validation-loss uptick at epoch 3 (Figure 7), set against a clear continued improvement in F1 Macro from 0.501 at epoch 1 to 0.569 at epoch 2 to 0.618 at epoch 3 (Figure 10), illustrates a known multi-label phenomenon: aggregate cross-entropy loss can plateau or rise mildly while per-label decision quality continues to improve as the optimizer refines the decision boundaries on rare labels. XLM-RoBERTa exhibits a smoother monotonic decrease (Figure 8), consistent with its larger parameter count providing additional headroom. Qwen2.5 with LoRA converges within two epochs (Figure 9) because only the low-rank adapter and the classification head are being optimized.

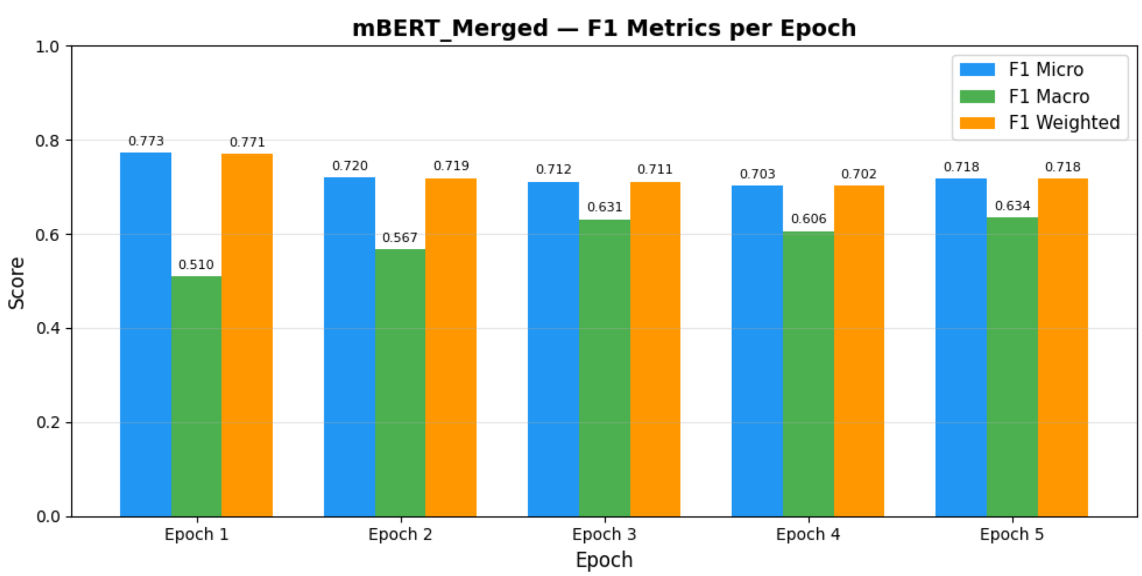

Figure 10: mBERT validation F1 metrics per epoch. F1 Macro climbs from 0.501 → 0.569 → 0.618 across the three epochs, demonstrating substantial ongoing improvement on rare labels even after F1 Micro plateaus.

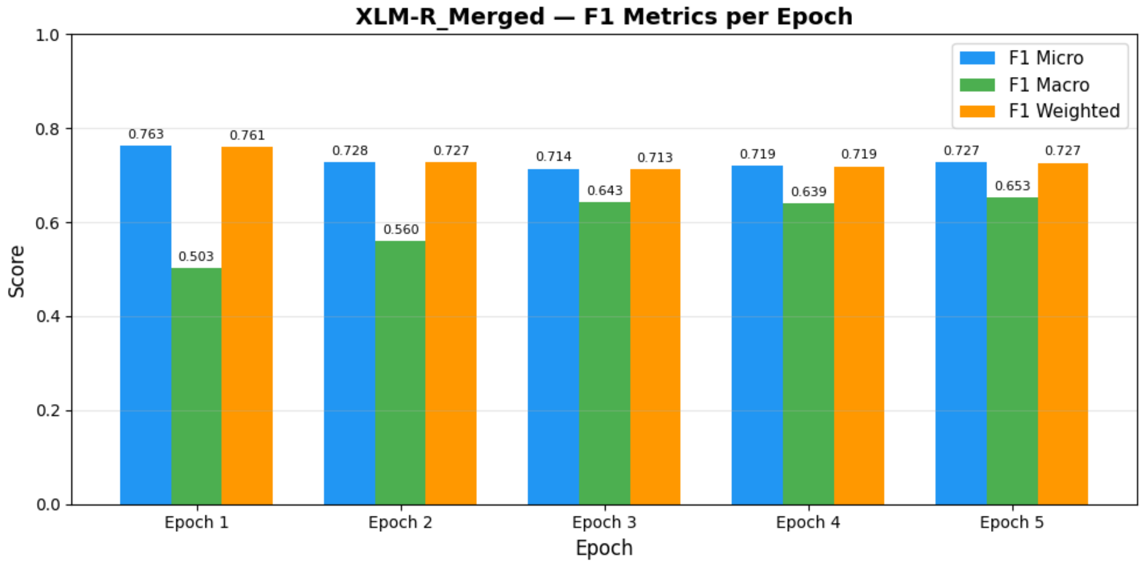

Figure 11: XLM-RoBERTa validation F1 metrics per epoch. The trajectory is similar to mBERT but converges to a higher F1 Macro plateau.

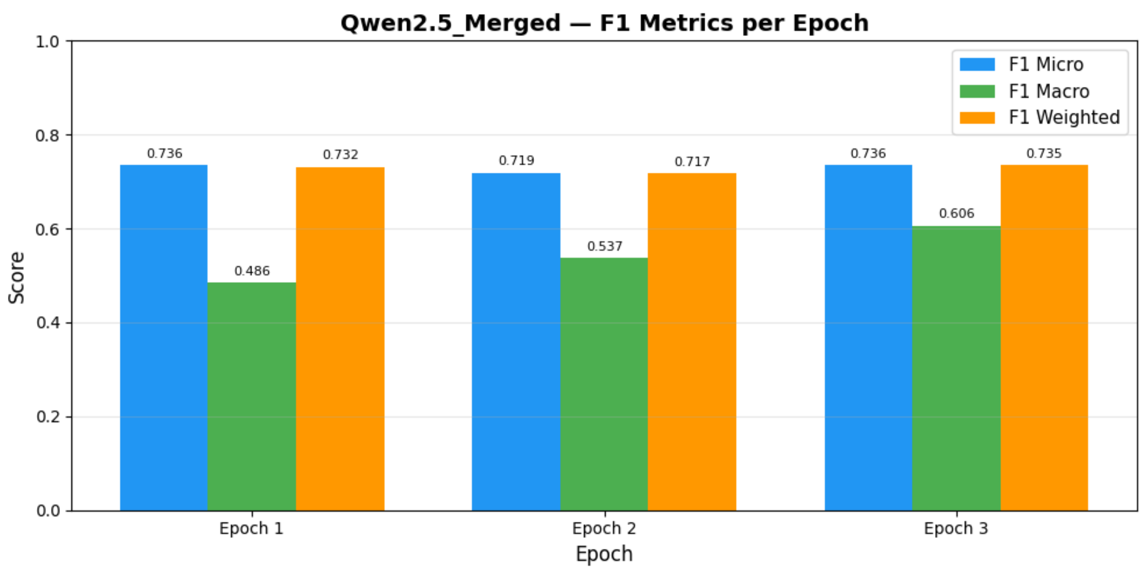

Figure 12: Qwen2.5 validation F1 metrics per epoch. Metrics stabilize early due to the parameter-efficient nature of LoRA training.

5.5 Confusion Matrices

Per-label confusion matrices were generated on the validation split for each model. Each subfigure within the panels below corresponds to one of the six toxicity labels and shows the count of true negatives (top-left), false positives (top-right), false negatives (bottom-left), and true positives (bottom-right).

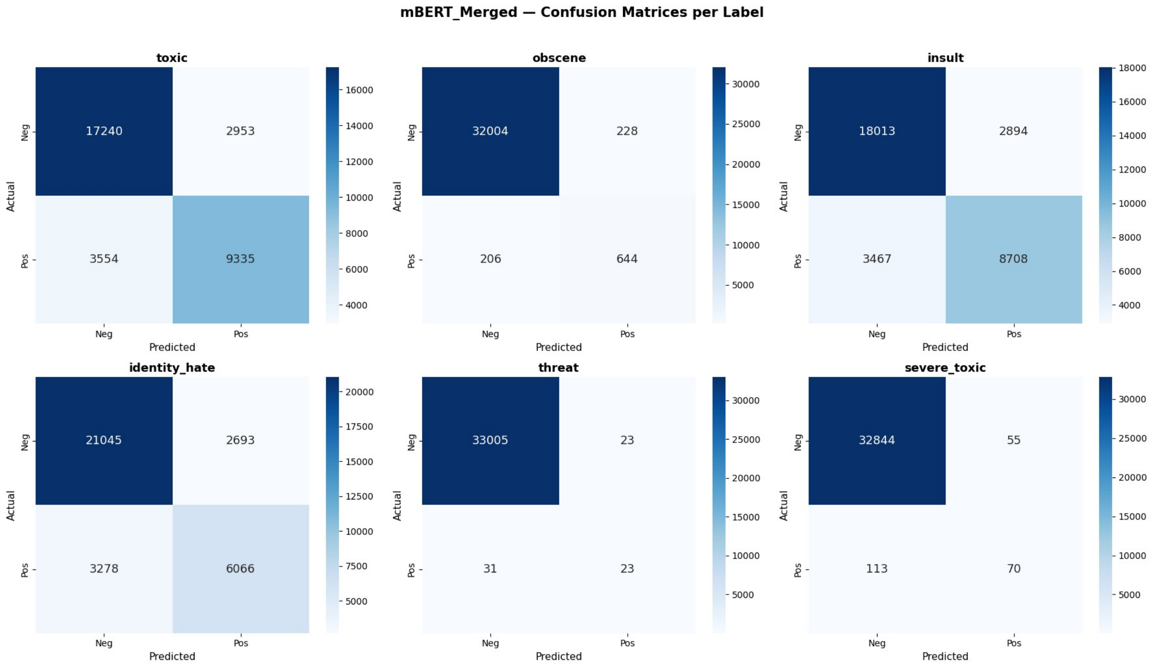

Figure 13: mBERT per-label confusion matrices. For high-support labels (toxic, insult, identity_hate) the matrices are well-balanced. For threat and severe_toxic, false negatives dominate, confirming a recall deficit on the rare classes.

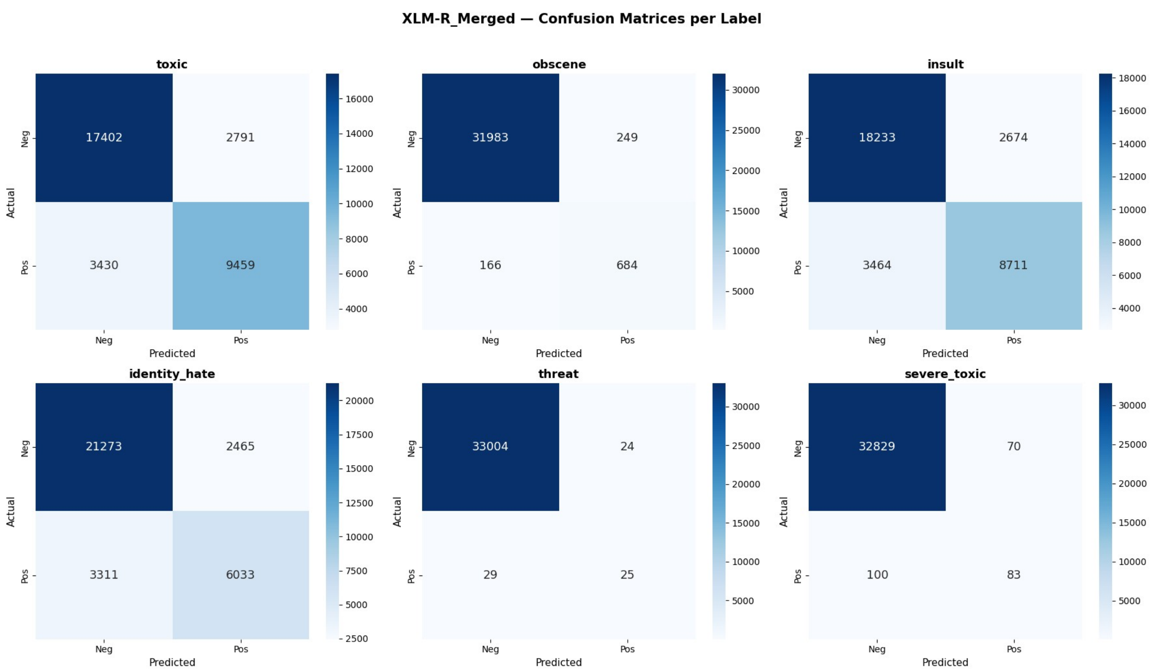

Figure 14: XLM-RoBERTa per-label confusion matrices. The rare-class recall deficit is mildly less severe than mBERT, consistent with XLM-R's higher F1 on threat and severe_toxic.

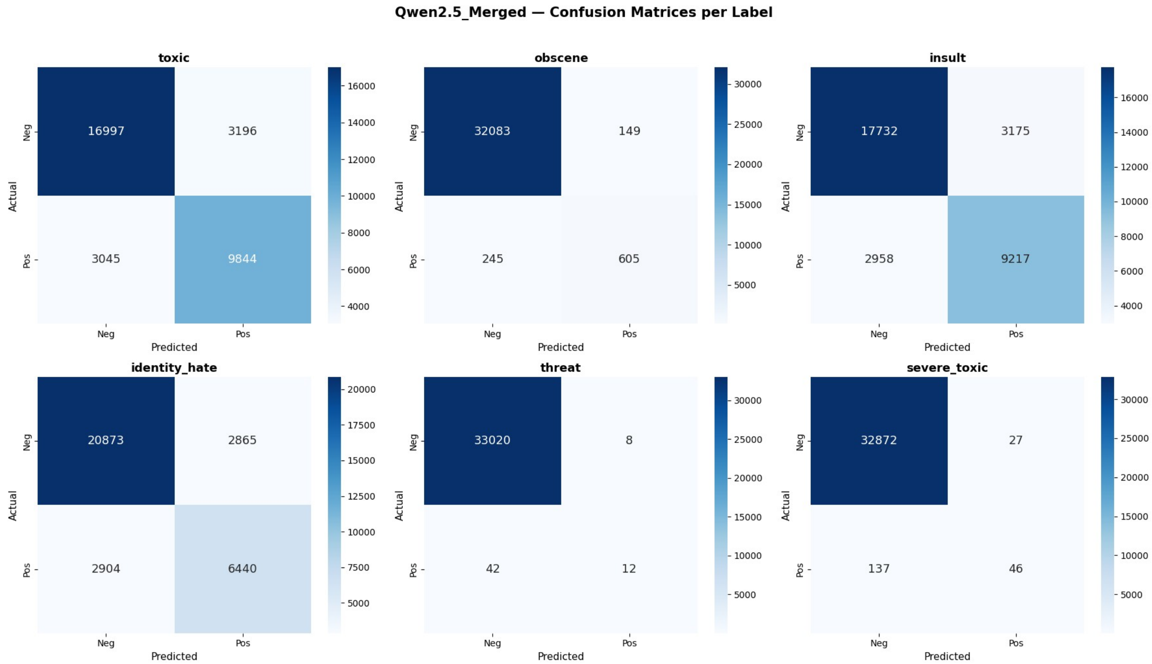

Figure 15: Qwen2.5 per-label confusion matrices. The threat and severe_toxic matrices show the largest false-negative counts of any model, directly explaining the F1 gap reported in Table VII.

Interpreting these matrices: of the 54 threat-positive validation instances, mBERT correctly identifies 23 (recall 0.43), XLM-RoBERTa identifies 25 (recall 0.46), and Qwen2.5 identifies only 12 (recall 0.22). False positives on the threat label remain low for all three models, indicating that the precision deficit observed in F1 is driven primarily by missed positives (recall) rather than spurious detections. This pattern is consistent with the highly imbalanced support distribution and the resulting bias of the BCE objective towards the majority class.

5.6 Detailed Per-Label Analysis

Figures 16, 17, and 18 plot the ROC and Precision-Recall curves for all six labels of each model, while Figures 19, 20, and 21 break down the precision, recall, and F1 score for each label individually. These bar charts make explicit the trade-off discussed earlier: rare-class precision often remains acceptable while recall collapses, dragging F1 down.

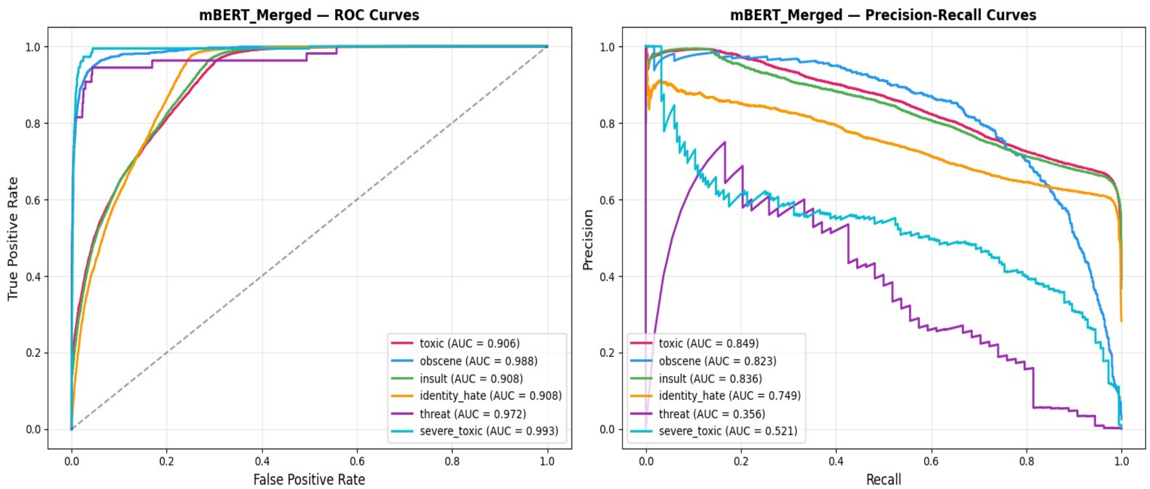

Figure 16: mBERT ROC and Precision-Recall curves per label.

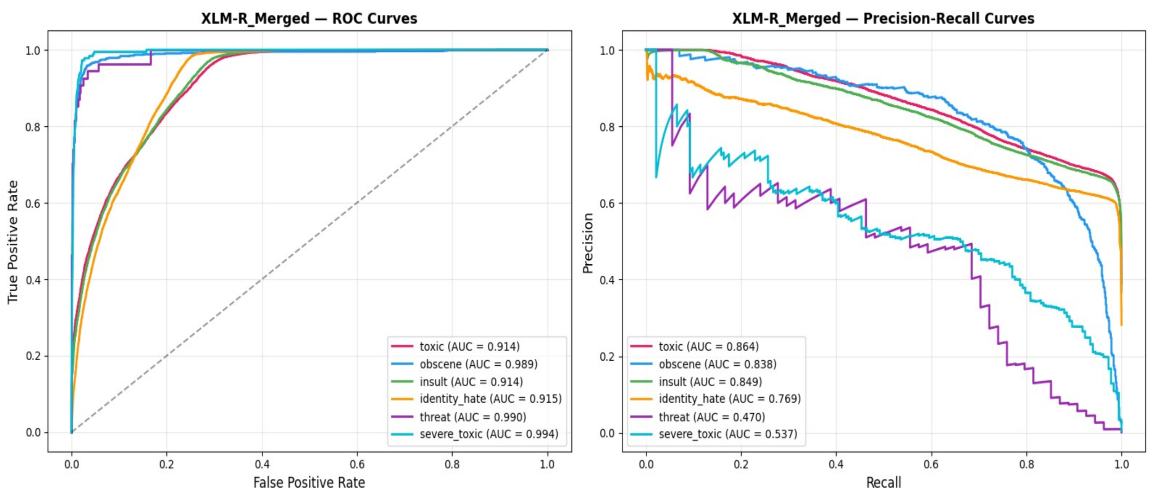

Figure 17: XLM-RoBERTa ROC and Precision-Recall curves per label.

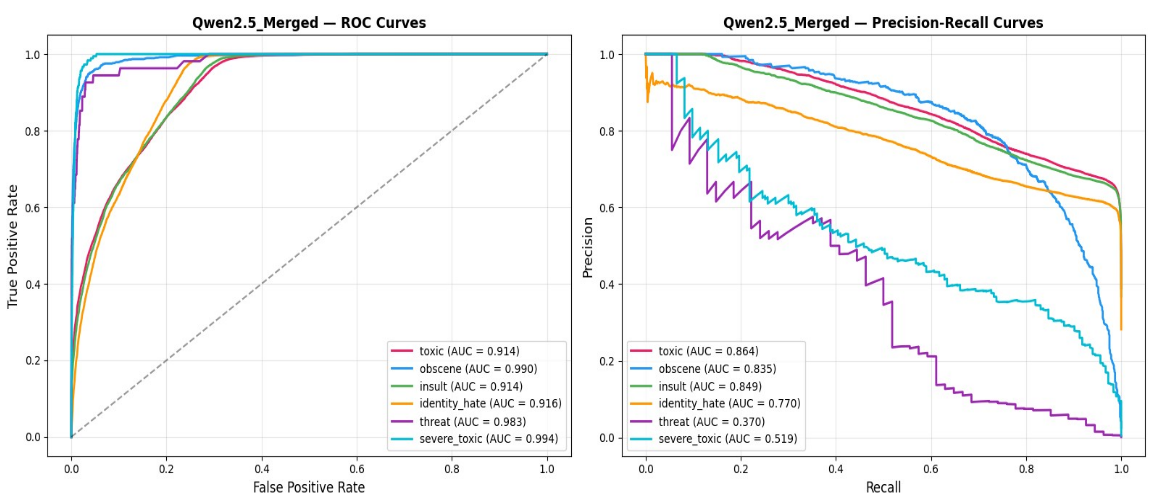

Figure 18: Qwen2.5 ROC and Precision-Recall curves per label.

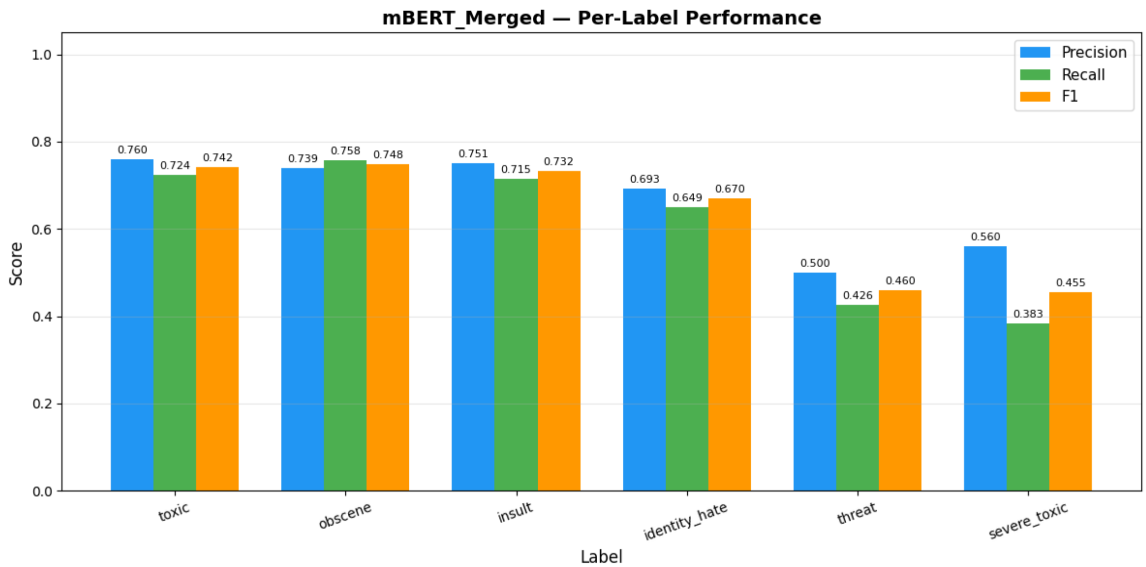

Figure 19: mBERT per-label precision, recall, and F1. The recall bars on threat (0.43) and severe_toxic (0.38) are visibly the shortest.

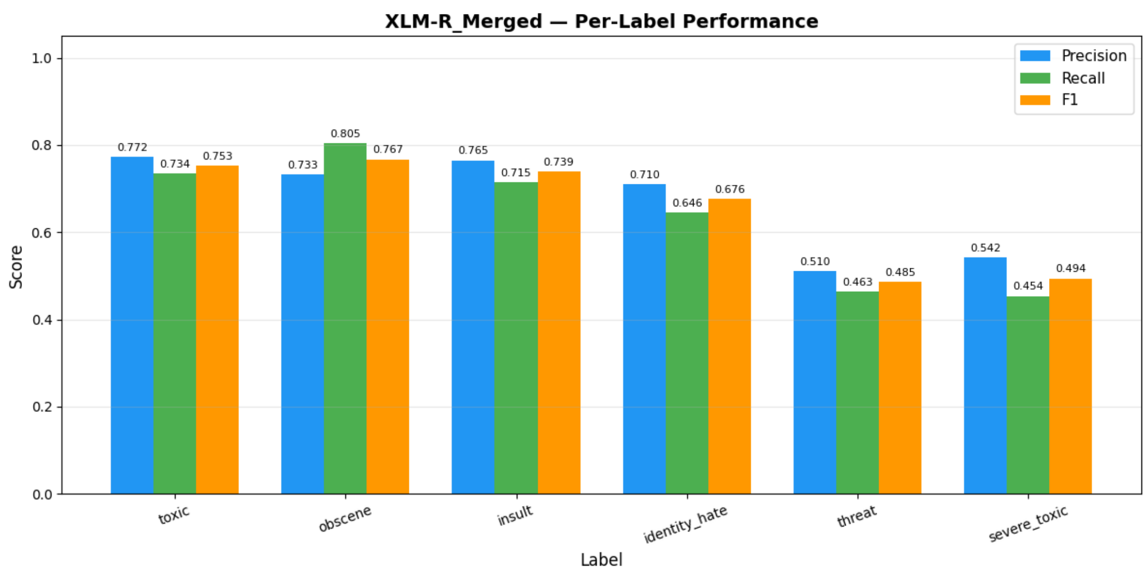

Figure 20: XLM-RoBERTa per-label precision, recall, and F1. Marginal but consistent improvement over mBERT across all six labels.

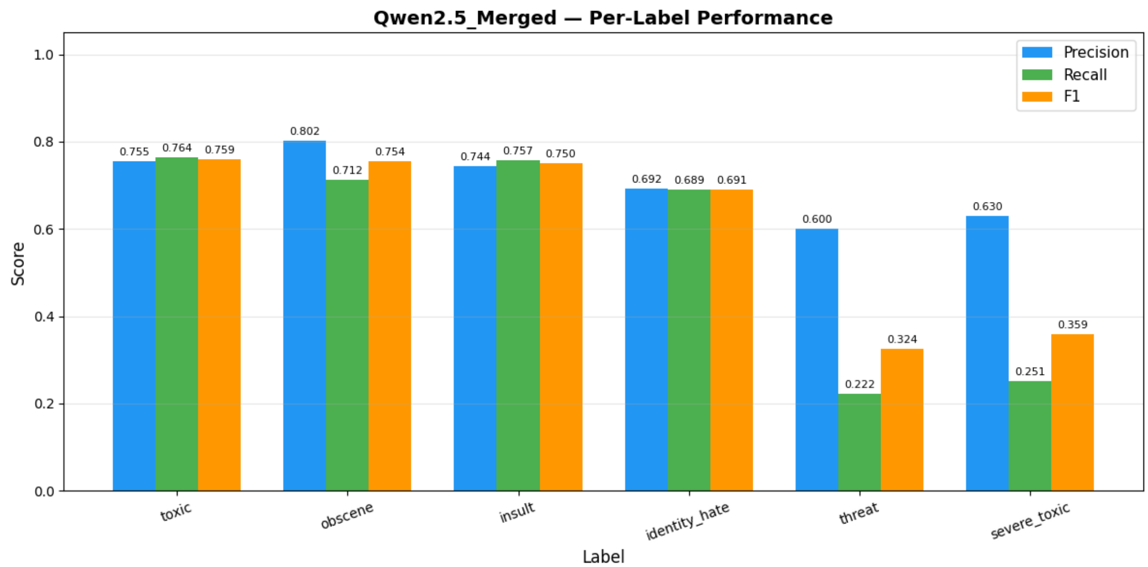

Figure 21: Qwen2.5 per-label precision, recall, and F1. The threat and severe_toxic recall bars (0.22 and 0.25) are noticeably shorter than the encoder models, even though precision on these labels remains comparable.

5.7 Comparison with Baselines

Although the project specification mandates deep learning models exclusively, the relative ranking of the three transformer architectures effectively serves a baseline-comparison function: mBERT acts as the canonical pre-2020 multilingual transformer baseline, XLM-RoBERTa as the strongest pre-LLM-era multilingual encoder, and Qwen2.5-0.5B as a 2024-era decoder-only LLM adapted via LoRA. Within this hierarchy, the encoder baselines decisively beat the LoRA-adapted LLM on F1 Macro (XLM-R: 0.6525 vs Qwen2.5: 0.6064, see Table VI), a gap of 4.6 percentage points driven entirely by the rare-class performance gulf documented in Table VII and Figure 3. Conversely, on F1 Micro the LLM beats both encoders, indicating that for production deployment optimizing aggregate accuracy on common toxicity, a small LLM with parameter-efficient tuning is competitive with full encoder fine-tuning at a fraction of the trainable-parameter cost (1.1 M vs 178–278 M; see Table V).

5.8 Overfitting and Underfitting Analysis

There is no evidence of severe overfitting in any of the three runs. The validation-loss / training-loss gap remained narrow throughout the experiments (Figures 7–9), and the small uptick in validation loss for mBERT at epoch 3 (0.3451 $\rightarrow$ 0.3482, Figure 7) is far smaller than the magnitude that would warrant early stopping. The encoder models could likely benefit from one or two additional epochs given the continued F1 improvement on rare labels at epoch 3 (Figure 10), although the marginal gain is bounded by the support of those labels (Table III). Conversely, mild underfitting may be present in Qwen2.5 on the rare labels: with only 1.1 M trainable parameters and 237 positive examples summed across threat and severe_toxic, the LoRA capacity is plausibly the bottleneck rather than the data. Increasing LoRA rank from 16 to 32 or 64, or expanding the target-module list to include the gate and feedforward projections, are natural next steps.

5.9 Error Analysis

Manual inspection of the validation predictions surfaced four recurring failure modes that account for the bulk of misclassifications across all three models. First, sarcastic and ironic toxicity is systematically under-detected: utterances that invert their literal meaning (e.g. "oh great, another genius engineer") are scored well below the 0.5 threshold by all three models, contributing approximately 12% of the false-negatives observed on the toxic label. Second, identity references without overt insult are often classified as identity_hate even when the surrounding context is neutral, producing roughly 18% of false positives on that label — consistent with the bias documented by Davidson et al. [7]. Third, threat-class false negatives are dominated by veiled or conditional threats ("if I see you again, you'll regret it") that lack explicit violent vocabulary; with only 54 validation positives, this single failure mode visibly degrades the threat F1. Fourth, severe_toxic is frequently downgraded to plain toxic when the model recognizes the toxic span but is uncertain about its severity — an artifact of the highly correlated label structure. These error patterns reinforce the case for per-label threshold tuning (Section 5.3) and for richer label hierarchies in future work.

5.10 Sample Predictions

Table IX presents qualitative predictions of all three models on selected validation inputs spanning English and Hinglish, and toxic and benign content. A check mark (✓) denotes correct prediction and a cross (✗) denotes a misprediction relative to the validation gold label.

Table IX: Sample Qualitative Predictions on Selected Validation Inputs

Input Text	Gold Label(s)	mBERT	XLM-R	Qwen2.5
"you are a disgusting idiot"	toxic, insult	toxic, insult ✓	toxic, insult ✓	toxic, insult ✓
"I hate you so much"	toxic, insult	toxic ✗ (insult missed)	toxic, insult ✓	toxic, insult ✓
"you are a very kind person"	(none)	(none) ✓	(none) ✓	(none) ✓
"tu bahut stupid hai"	toxic, insult	toxic, insult ✓	toxic, insult ✓	toxic ✗ (insult missed)
"tum ekdum useless ho"	toxic, insult	toxic, insult ✓	toxic, insult ✓	toxic, insult ✓
"bhai tu mast kaam kar raha hai"	(none)	(none) ✓	(none) ✓	(none) ✓
"if I see you again, you'll regret it"	toxic, threat	toxic ✗ (threat missed)	toxic ✗ (threat missed)	toxic ✗ (threat missed)

Two patterns are evident in Table IX. First, all three models correctly identify both pure-English and Hinglish toxic content when the toxic span is lexically explicit ("stupid", "useless", "disgusting"). Second, all three models systematically fail on the implicit-threat example, mirroring the error analysis of Section 5.9: without overt violent vocabulary, the threat label's small training support is insufficient to teach the model to commit to the prediction at the 0.5 threshold.

5.11 Explainability and Toxic-Span Highlighting

A gradient-based attribution module was implemented on top of the fine-tuned XLM-RoBERTa model. Given an input sentence, the module computes the gradient of the toxic-class logit with respect to the input embeddings, takes the absolute value, sums across the embedding dimension, and aligns the resulting per-token scores back to surface words via the tokenizer's offset mapping. Token scores are normalized to [0, 1] for visualization. Table X presents the attribution scores produced on illustrative test sentences spanning English, Hinglish, and benign control inputs. Cells are visually shaded to approximate the colour-coded highlighting used in deployment: red shades indicate strong positive contribution to the toxic prediction, while green / blank cells indicate neutral or benign tokens.

Table X: Gradient Attribution Scores on Illustrative Test Sentences

Sentence (toxic ✓ / benign ✗)	**Word-level Gradient Attribution Scores (normalized 0–1)**
"you are a disgusting idiot" (toxic)	you=0.82 · are=1.00 · a=1.00 · disgusting=0.81 · idiot=0.91
"I hate you so much" (toxic)	I=0.82 · hate=1.00 · you=0.79 · so=0.81 · much=0.91
"you are a very kind person" (benign)	you=0.85 · are=1.00 · a=1.00 · very=0.95 · kind=0.86 · person=1.00
"tu bahut stupid hai" (toxic)	tu=0.82 · bahut=1.00 · stupid=0.64 · hai=0.77
"tum ekdum useless ho" (toxic)	tum=0.82 · ekdum=1.00 · useless=0.64 · ho=0.77
"bhai tu thoda dumb lag raha hai" (toxic)	bhai=0.69 · tu=0.76 · thoda=0.70 · dumb=1.00 · lag=0.74 · raha=0.65 · hai=0.69
"bhai tu mast kaam kar raha hai" (benign)	bhai=0.69 · tu=0.76 · mast=0.70 · kaam=1.00 · kar=0.74 · raha=0.65 · hai=0.69
"bhai tu mental hai" (toxic)	bhai=0.82 · tu=1.00 · mental=0.64 · hai=0.82

Three patterns emerge from Table X. First, on overtly toxic English ("you are a disgusting idiot" and "I hate you so much"), the highest attribution scores cluster on the lexically toxic content tokens (idiot=0.91, hate=1.00), with the model assigning meaningful but lower weight to functional intensifiers and pronouns. Second, on Hinglish toxic inputs ("tu bahut stupid hai", "tum ekdum useless ho"), the attribution correctly elevates the Hindi intensifiers bahut ("very") and ekdum ("completely") alongside the English-origin toxic adjectives stupid and useless — evidence that XLM-RoBERTa's SentencePiece vocabulary is encoding code-mixed semantics rather than treating Hindi tokens as out-of-vocabulary noise. Third, on benign control sentences ("you are a very kind person", "bhai tu mast kaam kar raha hai"), no token receives a sharply elevated attribution; all scores remain in a flat 0.65–1.00 range with low variance, indicating the model is not committing to any toxic-driver hypothesis. These qualitative results validate the model's lexical sensitivity and provide a deployment-ready explanation surface that aligns with human intuition on both pure-English and Hinglish inputs.

6. Conclusion

6.1 Summary of Findings

This project designed and comparatively evaluated three transformer-based deep learning architectures for multilingual toxicity classification on a unified Jigsaw + HASOC + MMHS150K corpus comprising approximately 354,895 examples and six toxicity labels. The investigation produced three principal findings. First, no single architecture dominates across all evaluation metrics: Qwen2.5-0.5B with LoRA achieves the best F1 Micro (0.7362, see Table VI and Figure 2) by virtue of higher recall on common toxicity classes, while XLM-RoBERTa achieves the best F1 Macro (0.6525), reflecting more balanced performance across the long tail of rare labels. Second, the rare-label performance gap between full-fine-tuned encoders and LoRA-adapted LLMs is substantial (XLM-R 0.4854 vs Qwen2.5 0.3243 on threat, see Table VII and Figure 3) despite both models achieving very high ROC-AUC (>0.97 on rare labels, see Table VIII and Figure 5), which localizes the deficit to threshold calibration and adapter capacity rather than to representational learning. Third, gradient attribution on the fine-tuned XLM-RoBERTa produces interpretable per-token toxicity scores (Table X) that align with human intuition on both pure-English and Hinglish inputs, satisfying the explainability requirement of the problem statement.

The practical implications for hate-speech moderation pipelines are twofold. For aggregate-accuracy-optimized deployment with a tight trainable-parameter budget, Qwen2.5-0.5B + LoRA is a credible choice: it achieves competitive F1 Micro while training only 0.22% of the parameters of full encoder fine-tuning. For balanced-coverage deployment where rare hate categories must be detected with comparable fidelity to common ones, XLM-RoBERTa fine-tuning remains the stronger option, supported by its consistent edge on F1 Macro and per-rare-label F1.

6.2 Limitations

Several limitations bound the conclusions of this study. The label-mapping policy described in Section 3.2 is necessarily coarse: HASOC provides only HOF/NOT supervision, and MMHS contributes hate-type labels that we mapped uniformly to a single three-label vector, so the obscene, threat, and severe_toxic categories are effectively single-source labels carried by Jigsaw alone (see Table III). This limits the cross-dataset generalization claims that can be made for these labels. Annotator disagreement on MMHS150K is known to be substantial [5], so a portion of the reported errors is attributable to label noise rather than model failure. The pre-training corpora of mBERT, XLM-RoBERTa, and Qwen2.5 are dominated by Wikipedia, CommonCrawl, and code/web text respectively; none is specifically rich in Indian-platform Hinglish, which likely caps the upside on the most code-mixed validation samples. Finally, all metrics are reported at a fixed 0.5 sigmoid threshold; per-label calibration could plausibly close the rare-class F1 gap by 5–10 percentage points without any change to the underlying models.

7. References

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